Estimating Rupture Dimensions of Three Major Earthquakes in Sichuan, China, for Early Warning and Rapid Loss Estimates

Jiawei Li[1,2,3], Maren Böse*[3], Max Wyss[4], David J. Wald[5], Alexandra Hutchison[3,6], John F. Clinton[3], Zhongliang Wu[7], Changsheng Jiang[1], and Shiyong Zhou[2]

ABSTRACT

Large earthquakes, such as Wenchuan in 2008, M_w 7.9, Sichuan, China, provide an opportunity for earthquake early warning (EEW), as many heavily shaken areas are far (~ 50 km) from the epicenter and warning times could be sufficient (≥ 5 s) to take preventive action. On the other hand, earthquakes with magnitudes larger than ~ M 6.5 are challenging for EEW because source dimensions need to be defined to adequately estimate shaking. Finite-fault rupture detector (FinDer) is an approach to identify fault rupture extents from real-time seismic records. In this study, we playback local and regional onscale strong-motion waveforms of the 2008 M_w 7.9 Wenchuan, 2013 M_w 6.6 Lushan, and 2017 M_w 6.5 Jiuzhaigou earthquakes to study the performance of FinDer for the current layout of the China Strong Motion Network. Overall, the FinDer line-source models agree well with the observed spatial distribution of aftershocks and models determined from waveform inversion. However, because FinDer models are constructed to characterize seismic ground motions (as needed for EEW) instead of source parameters, the rupture length can be overestimated for events radiating high levels of high-frequency motions. If the strong-motion data used had been available in real time, 50%–80% of sites experiencing intensity modified Mercalli intensity IV–VII (light to very strong) and 30% experiencing VIII–IX (severe to violent) could have been issued a warning with 10 and 5 s, respectively, before the arrival of the S wave. We also show that loss estimates based on the FinDer line source are more accurate compared to point-source models. For the Wenchuan earthquake, for example, they predict a four to six times larger number of fatalities and injured, which is consistent with official reports. These losses could be provided 1/2 ~ 3 hr faster than if they were based on more complex inversion rupture models.

KEY POINTS

- Rapid finite-source models of earthquakes are important for early warning and loss estimates.
- We compute finite-fault models of three major earthquakes in Sichuan, China, using the FinDer algorithm.
- Solutions are in good agreement with aftershock distributions and inversion models, but are computed faster.

Supplemental Material

INTRODUCTION

Seismic risk in parts of China is high: although accounting for only a quarter of the world's land surface, about one-third of the world's most devastating onshore earthquakes in the twentieth century occurred in China (Ma and Zhao, 1991). An $M_s > 7.5$ earthquake hits China on average every five years, an $M_s > 8.0$ every 10 yr, based on nearly 100 yr of statistical data (Department of Earthquake Disaster Prevention of the State Seismological Bureau, 1995; Department of Earthquake Disaster Prevention of the China Earthquake Administration, 1999; Gao *et al.*, 2014). Sichuan province has been hit particularly hard in the recent past with the occurrences of the 2008 M_w 7.9 Wenchuan, 2013 M_w 6.6 Lushan, and 2017 M_w 6.5

1. Institute of Geophysics, China Earthquake Administration, Beijing, China; 2. School of Earth and Space Sciences, Peking University, Beijing, China; 3. Swiss Seismological Service, Swiss Federal Institute of Technology Zürich (ETH Zürich), Zürich, Switzerland; 4. International Centre for Earth Simulation Foundation, Geneva, Switzerland; 5. U.S. Geological Survey, Denver, Colorado, U.S.A.; 6. Institut français des sciences et technologies des transports, de l'aménagement et des réseaux (IFSTTAR), Paris, France; 7. Institute of Earthquake Forecasting, China Earthquake Administration, Beijing, China

*Corresponding author: mboese@sed.ethz.ch

Cite this article as Li, J., M. Böse, M. Wyss, D. J. Wald, A. Hutchison, J. F. Clinton, Z. Wu, C. Jiang, and S. Zhou (2020). Estimating Rupture Dimensions of Three Major Earthquakes in Sichuan, China, for Early Warning and Rapid Loss Estimates, *Bull. Seismol. Soc. Am.* **110**, 920–936, doi: 10.1785/0120190117

Jiuzhaigou earthquakes, causing in total more than 87,000 fatalities, 388,600 injured, and economic losses of greater than \$1.5 trillion U.S. (Chen and Booth, 2011; Wu *et al.*, 2014; Han *et al.*, 2018).

Shortly after the devastating M_w 7.9 Wenchuan earthquake, seismologists recommended that China implements an earthquake early warning (EEW) system, following the examples of other nations (Wu *et al.*, 2008; Wan *et al.*, 2009). EEW is the effort to rapidly detect and characterize potentially damaging earthquakes and to issue warnings to areas before they experience strong shaking (e.g., Allen *et al.*, 2009; Satriano, Wu, *et al.*, 2011; Allen and Melgar, 2019). EEW is considered a potentially effective component of seismic risk reduction (Strauss and Allen, 2016). Several countries and regions are testing EEW algorithms in their seismic monitoring networks (Chile, Switzerland, Italy, and central America; Zollo *et al.*, 2009; Satriano, Elia, *et al.*, 2011; Behr *et al.*, 2015; Clinton *et al.*, 2016; Leyton *et al.*, 2018; Massin *et al.*, 2019) or have fully operational systems to alert end users (Romania, Turkey, and Korea; Erdik *et al.*, 2003; Böse *et al.*, 2007; Ionescu *et al.*, 2007; Sesetyan *et al.*, 2011; Sheen *et al.*, 2016) or the general public (Japan, Mexico, United States, and Taiwan; Espinosa-Aranda *et al.*, 1995; Wu and Teng, 2002; Horiuchi *et al.*, 2005; Allen, 2007; Wu *et al.*, 2007; Kamigaichi *et al.*, 2009; Given *et al.*, 2018).

In 2007, prior to the Wenchuan earthquake, the Institute of Geophysics (IGP) at the China Earthquake Administration (CEA) and the Department of Geosciences at National Taiwan University had built an EEW prototype system for the Beijing capital region (BCR) (Peng *et al.*, 2011). This system was based on the Capital Circle Seismograph Network of China and included 94 broadband and 68 short-period stations, with an average interstation spacing of ~50 km. Following the Wenchuan earthquake a number of research studies on EEW in China were conducted, the prototype test system in BCR was expanded (Peng *et al.*, 2011) and a new test system in Fujian province of China was built (Zhang *et al.*, 2016).

Within the scope of the National System for Fast Report of Intensities and EEW project (Zhang *et al.*, 2016; Zheng, 2018), led by the CEA, a nationwide EEW system covering the four main seismic zones (BCR, southeastern coastal area, northern Xinjiang area, and central China north–south seismic belt) is currently being built. During the 13th five-year plan period (2016–2020), close to ~2000 broadband stations (equipped with three-component broadband or very broadband seismometers and all with accelerometers), ~3200 strong-motion stations (equipped with three-component accelerometers), and ~10, 200 low-cost intensity sensors (equipped with microelectromechanical systems; Peng, Chen, *et al.*, 2017) will be deployed or upgraded throughout mainland China by 2020. This system will build on the Jopens seismic monitoring platform developed by CEA (Chaoyong Peng, written comm., 2019).

The destructive 2008 Wenchuan, 2013 Lushan, and 2017 Jiuzhaigou earthquakes hit after China's mostly digital broadband and strong-motion networks had been established. These networks provide critical data for retrospective studies of the feasibility of EEW in Sichuan province. Wang *et al.* (2009), Ma *et al.* (2011), Peng *et al.* (2014), Peng, Yang, *et al.* (2017), and Li *et al.* (2017) studied the regression relationships between various frequency and amplitude measures (such as τ_p^{max}, τ_c, and P_d), and earthquake magnitude in different time windows around the *P*-wave arrival for the Wenchuan main- and aftershocks. Wang and Zhao (2018) conducted a similar analysis using the squared integral displacement (ID2). Yamada (2014) proposed a function to identify the Wenchuan fault rupture extent by classifying stations into near- and far-source observations. Wang *et al.* (2017) proposed a real-time numerical shake prediction and updating method to predict ground motion. For the same event, Fang *et al.* (2014) investigated the application of high-rate Global Positioning System (GPS) observations to invert in (near) real time for seismic extended source parameters. Additional EEW studies were conducted for the Lushan earthquake (Li *et al.*, 2017; Peng, Yang, *et al.*, 2017), but no or little EEW research has yet been done for the Jiuzhaigou earthquake, mainly because of its recent occurrence and its poor station coverage.

The M_w 7.9 Wenchuan earthquake is one of the largest continental earthquakes ever recorded within a high-quality seismic network. It generated a 300 km long quasi-unilateral rupture along the Longmenshan fault zone and resulted in destructive ground motions over an area of about 450, 000 km^2 (Chen and Booth, 2011). Large earthquakes like this event are particularly challenging for EEW, because source characterizations are more complex: first, seismic records tend to saturate on traditional high-gain seismic network stations, which lead to magnitude underestimations, and, second, source dimensions need to be known to adequately predict seismic ground motions, which are controlled primarily by the distance to the rupturing fault. Both magnitude underestimation and neglecting of source dimensions contributed to the underestimation of ground motions in the Japanese public EEW system operated by the Japan Meteorological Agency (JMA) during the 2011 M_w 9.0 Tohoku–Oki subduction zone earthquake (Hoshiba and Aoki, 2015). Therefore, estimating fault rupture extent and orientation, as well as focal mechanisms in real time are important and are fields of active research (Böse *et al.*, 2012, 2015, 2018; Zhang, Wang, Zschau, *et al.*, 2014; Nazeri *et al.*, 2019; Tarantino *et al.*, 2019).

In this study, we playback strong-motion waveforms of the Wenchuan, Lushan, and Jiuzhaigou earthquakes to simulate the performance of the finite-fault rupture detector (FinDer) algorithm (Böse *et al.*, 2012, 2015, 2018) for the current layout of the China Strong Motion Network. Using the example of the Wenchuan earthquake, we also discuss the potential application of FinDer line-source models for rapid loss estimates in the immediate aftermath of significant earthquakes. The strong-motion data used in this retrospective study were unavailable in real time, when these three earthquakes hit. However, an increasing number of stations are being added to the National System for Fast Report of Intensities and EEW and can contribute to future alerts.

FINDER

Estimating the dimensions of earthquake fault ruptures in (near) real time is critical for EEW and rapid response, because seismic ground motions and damage tend to be largest at sites close to the rupture, which may extend far from the epicenter. In this study, we apply the FinDer algorithm (Böse *et al.*, 2012, 2015, 2018) to estimate rupture dimensions from seismic data.

FinDer calculates line-source models of earthquakes by matching the spatial distribution of high-frequency ground-motion amplitudes observed in a seismic network with theoretical template maps. These templates are computed from empirical ground-motion prediction equations (GMPEs) for line sources of different lengths and magnitudes. The strike of the fault rupture is determined by rotating the templates around various trial angles and calculating the respective misfit with the observed data. Together with the optimum spatial position and orientation, the template with the highest correlation and smallest misfit with the observed amplitudes is found from a combined grid search and divide-and-conquer approach (Böse *et al.*, 2018). The FinDer algorithm is computationally highly efficient and allows updating rupture dimensions every second until peak shaking is reached across the seismic network.

In this study, we construct FinDer templates $T(x, y|L)$ from GMPEs for peak ground acceleration (PGA) after Cua and Heaton (2009) as

$$\begin{aligned} T(x, y|L) &= \log_{10}(\mathrm{PGA}(x, y)) \\ &= aM - b[\sqrt{R^2 + 9} + C(M)] \\ &\quad + d \log_{10}[\sqrt{R^2 + 9} + C(M)] + e \end{aligned} \tag{1}$$

with

$$C(M) = c_1 e^{c_2(M-5)} \times \left[\arctan(M - 5) + \frac{\pi}{2}\right],$$

in which distance R is given in kilometers and PGA in $\mathrm{cm/s^2}$. R is either the epicentral distance if magnitude M is smaller than 5 or the distance to the rupture for larger events (this is $R(L, \theta)$). The soil conditions of the stations were defaulted as rock, and the corresponding regression coefficients (a, b, c_1, c_2, d, and e) in table 3 of Cua and Heaton (2009) are applied to equation (1) to calculate FinDer templates. For prediction of PGA at later times and/or at larger distances, we apply the same relationship, while using the shortest distance to the FinDer line source as distance metric. Magnitudes are estimated from the empirical rupture length to magnitude relation of Wells and Coppersmith (1994) (for all types of earthquakes) using the FinDer-estimated rupture length L,

$$M = 1.49 \log_{10}(L) + 4.38. \tag{2}$$

Along with the EPIC (Earthquake Point-Source Integrated Code) algorithm (Chung *et al.*, 2019), FinDer contributes to the U.S. west coast ShakeAlert warning system (Given *et al.*, 2018), currently in phase 2 of a public rollout in California, Washington, and Oregon. FinDer is undergoing additional real-time testing in central America (Nicaragua, Costa Rica, and El Salvador), Chile, and Switzerland (Böse *et al.*, 2018).

EVENTS AND AVAILABLE DATASETS

The Wenchuan, Lushan, and Jiuzhaigou earthquakes

The 12 May 2008 M_w 7.9/M_s 8.0 Wenchuan earthquake that occurred at 14:28:04 local time (103.42° E, 31.01° N, China Earthquake Network Center [CENC]; Fig. 1, Table 1), is one of the most devastating earthquakes worldwide. It generated PGA amplitudes of close to 1*g* at the nearest strong-motion station (WCW station; Fig. 2a) and a seismic intensity of up to XI on both the Chinese and the modified Mercalli intensity (MMI) scale (Chen and Booth, 2011). The Wenchuan oblique-thrust earthquake ruptured along two parallel subfaults in the Longmenshan fault zone of ~240 km and ~90 km in length, respectively (Chen and Booth, 2011). The highest intensities were observed in Wenchuan and Beichuan Counties, apparently linked to the rupture of two separate large asperities (Fig. 2a). As of 25 August 2008, 87,149 people were reported dead (including 17,923 missing) and at least 374,643 were injured (Chen and Booth, 2011). The direct economic loss was estimated as 845.1 billion Yuan/RMB ($1.5 trillion U.S.; Chen and Booth, 2011).

About 5 yr later, on 20 April 2013, at 08:02:46 local time, the M_w 6.6/M_s 7.0 Lushan earthquake (102.99° E, 30.3° N; published by CENC, Fig. 1, Table 1) struck in the southern part of the Longmenshan fault zone, only 90 km southwest of the Wenchuan epicenter. The focal mechanism solutions published by various agencies, including CEA, CENC, the Global Centroid Moment Tensor, JMA, and the U.S. Geological Survey (USGS), consistently indicated a thrust-dominated rupture (strike 218°, dip 39°; Wu *et al.*, 2014). Although aftershocks were mainly distributed on both sides of the Shuangshi–Dachuan fault, the seismogenic fault of this earthquake may have been a blind thrust fault in the eastern part of the Shuangshi–Dachuan fault zone. As of 24 April 2013, the State Council of China reported 217 fatalities (including 21 missing) and at least 13,484 injured (Wu *et al.*, 2014).

On 8 August 2017 at 21:19:49 local time, nine years after the 2008 Wenchuan and four years after the 2013 Lushan earthquake, the M_w 6.5/M_s 7.0 Jiuzhaigou earthquake (103.82° E, 33.2° N; published by CENC; Fig. 1, Table 1) hit ~150 km from the northeastern end of the Wenchuan rupture in the very

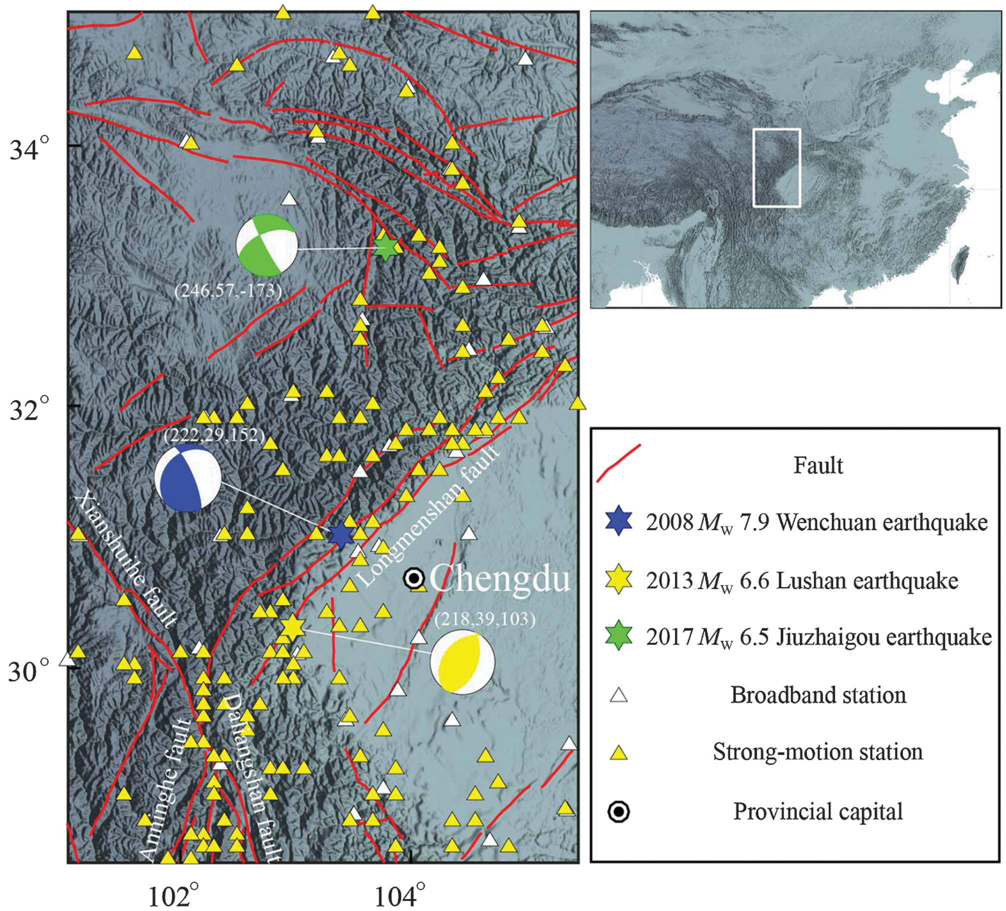


Figure 1. Epicenters (China Earthquake Network Center [CENC]) and focal mechanisms (U.S. Geological Survey [USGS]) of the 12 May 2008 M_w 7.9 Wenchuan, the 20 April 2013 M_w 6.6 Lushan, and the 8 August 2017 M_w 6.5 Jiuzhaigou earthquakes in Sichuan province, China. Map in the upper right shows the larger tectonic context with the Tibetan plateau visible from topography. Sichuan province is covered by a dense seismic network of broadband and strong-motion instruments (here as of 2018). Some new stations have been added to the networks after the Wenchuan, Lushan, and Jiuzhaigou earthquakes. Fault locations are taken from Deng *et al.* (2003).

northeastern corner of the Bayanhar block (Han *et al.*, 2018). The identification of the fault rupturing in the Jiuzhaigou earthquake is difficult, because no surface rupture was observed. The focal mechanism indicates that the rupture was dominated by strike slip along with thrusting non-double-couple components on a nearly vertical fault (Han *et al.*, 2018; Sun *et al.*, 2018; Zhang *et al.*, 2018; Yong Zhang, written comm., 2018). This earthquake caused 25 fatalities, six people missing, and 525 injured.

Waveform data

To simulate the real-time performance of FinDer for the Wenchuan, Lushan, and Jiuzhaigou earthquakes, we run playbacks of strong-motion waveforms obtained from the China Strong Motion Networks Center (CSMNC) at the Institute of Engineering Mechanics, CEA (see Data and Resources). CSMNC operates the China Strong Motion Networks, which started their official operation in March 2008, only a few months before the Wenchuan earthquake hit (Li *et al.*, 2008). The CSMNC is responsible for the technical support, management, and maintenance of the strong-motion networks, as well as for the collection, processing, and dissemination of acceleration data. CSMNC strong-motion stations are mainly distributed in 21 National Significant Seismic Monitoring and Protection Regions of mainland China. The network is constantly growing, mainly in response to the destructive earthquakes studied in this article.

TABLE 1
Catalog and Final FinDer Line-Source Parameters for the 2008 Wenchuan, 2013 Lushan, and 2017 Jiuzhaigou Earthquakes

		Epicenter*					Final FinDer Line-Source Model		
Earthquake	**Origin Time (yyyy/mm/dd) (Local Time)***	**Latitude (°)**	**Longitude (°)**	**Z (km)**	**M_w†/M_s***	**Strike/Dip/Rake†**	**L (km)**	**Θ (°)**	**M_{FD}**
Wenchuan	2008/05/12 14:28:04	31.01	103.42	16	7.9/8.0	222/29/152	290	230	8.0
Lushan	2013/04/20 08:02:46	30.3	102.99	17	6.6/7.0	218/39/103	99	215	7.3
Jiuzhaigou	2017/08/08 21:19:49	33.2	103.82	10	6.5/7.0	153/84/–33	33	130	6.6

FinDer, Finite-fault rupture detector; L, rupture length; Z, source depth; Θ, rupture strike.
*CENC, China Earthquake Network Center.
†USGS, U.S. Geological Survey.

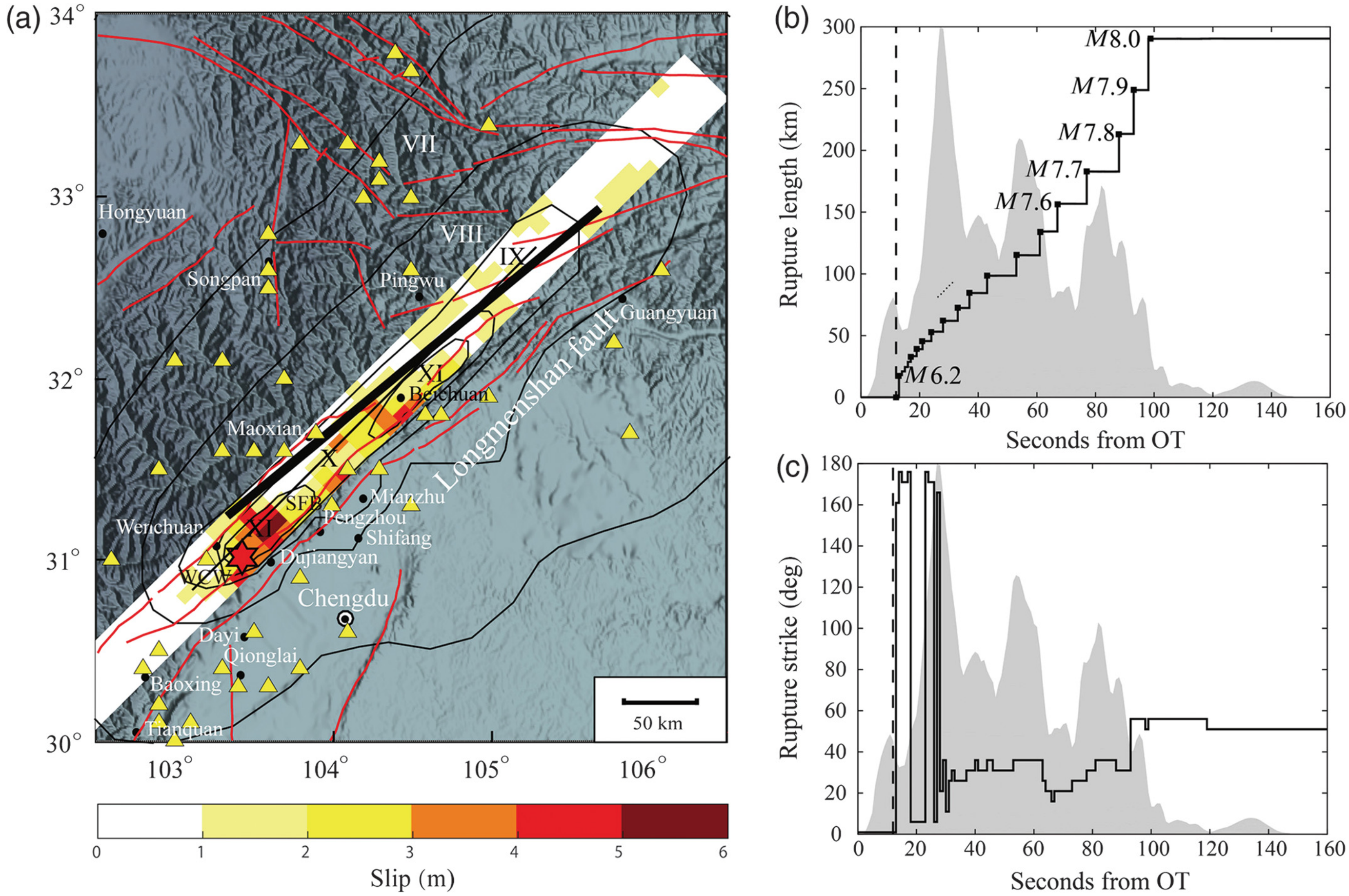


Figure 2. Setting of the 2008 M_w 7.9 Wenchuan earthquake and playback results. (a) Final finite-fault rupture detector (FinDer) line-source (black line; $L \sim 290$ km, $\Theta = 50°$, M_{FD} 8.0) and slip model after Zhang, Wang, Zschau, *et al.* (2014), which dips at 29° toward the northwest. Map shows only stations used in FinDer. The thin black lines show the isoseismals released by the China Earthquake Administration (CEA) (Chen and Booth, 2011). (b) Temporal evolution of FinDer rupture length L and magnitude M_{FD} from event origin time (OT). (c) Temporal evolution of FinDer estimated rupture strike Θ. Dashed lines in (b) and (c) mark FinDer trigger time (requires exceedance of 20 cm/s^2 at two or more stations). Source time function after Zhang, Wang, Zschau, *et al.* (2014) is shown in gray. Other features follow Figure 1.

In this study, we focus on Sichuan province, covered by a dense seismic network of broadband and strong-motion instruments deployed along the Longmenshan, Xianshuihe, Anninghe, Zemuhe, and Daliangshan faults (Fig. 1). Seismically active faults and seismicity in the Sichuan–Yunnan region result from the collision of the Eurasian and India plates, forming the Himalayan mountain zone. The index map in the upper right of Figure 1 shows the larger tectonic context with the Tibetan plateau visible.

A large number of high-gain broadband waveforms recorded during the 2008 Wenchuan earthquake and some of the near-source records of the 2013 Lushan and 2017 Jiuzhaigou earthquakes clipped. Therefore, we use in this study only strong-motion acceleration waveforms. In fact, for large rupture events, which are most relevant for EEW, the broadband stations will be always saturated in areas where damage can be expected. In real-time operation of FinDer, stations with clipped amplitudes are automatically discarded. However, for smaller earthquakes ($M < 6.5$), in which broadband records do not clip, these records can still serve as an important supplement in cases in which strong-motion records are insufficient.

All records obtained from CSMNC start 20 s before the P-wave arrival. For preprocessing, we conduct a baseline correction of each trace by subtracting the mean noise amplitude taken over the first 15 s. Absolute times for the records are not available, because the GPS clocks of some stations failed and seismic engineering has in general no need for this information (Liangqiong Lou, written comm., 2019). In this study, we reconstruct absolute times by taking the event locations and origin time (OT) from the CENC earthquake catalog and align the theoretical P-wave arrival (assuming $V_P = 6.0$ km/s) with the P arrival in each record. Finally, we determine the peak absolute ground-motion amplitude taken over all three waveform components in time windows of 1 s and feed their logarithmic value into FinDer (see Böse *et al.*, 2018, for details). FinDer uses the spatial distribution of PGA, including the information that certain stations have not yet recorded strong

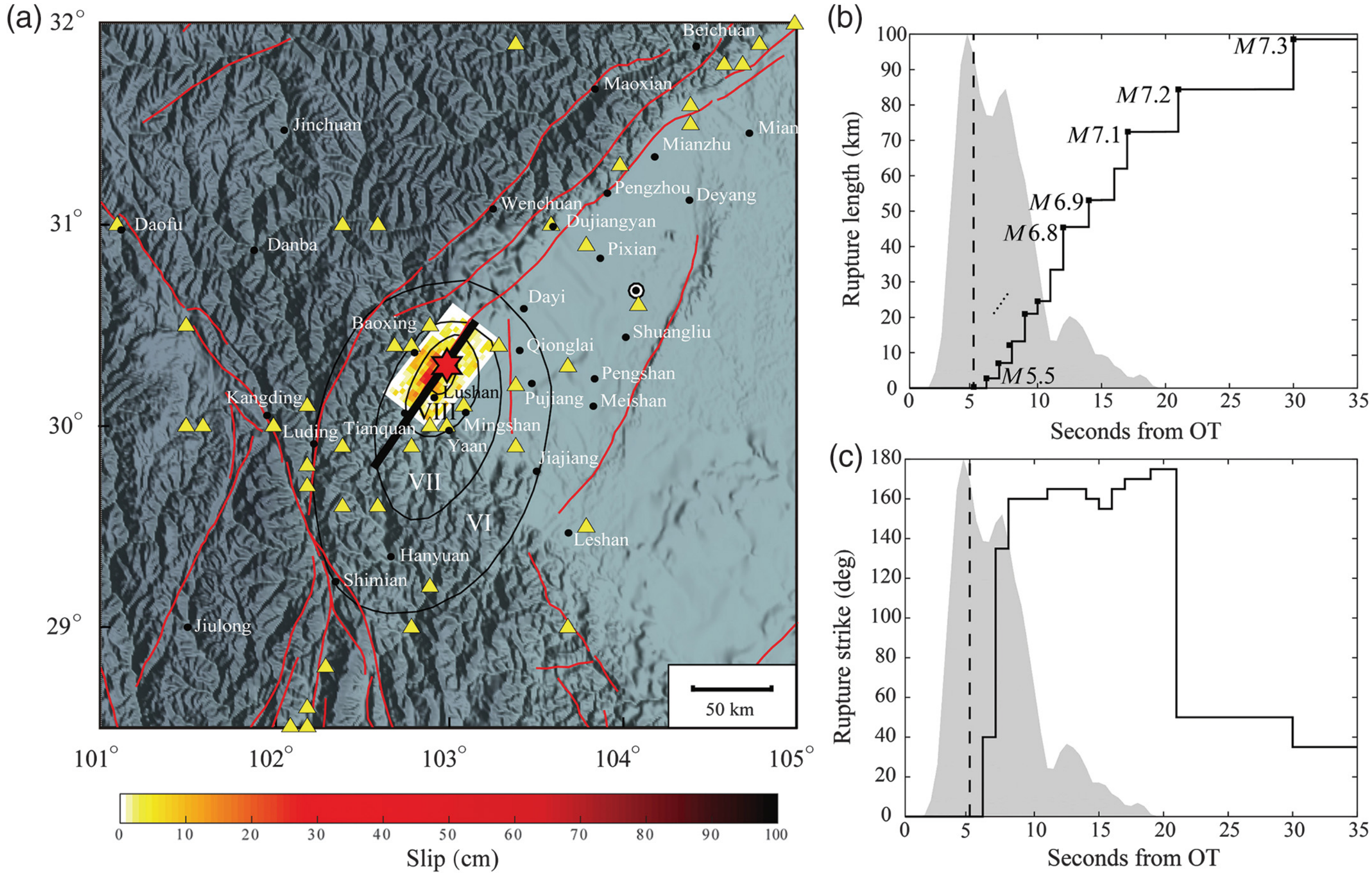


Figure 3. Setting of the 2013 M_w 6.6 Lushan earthquake and playback results (follows Fig. 2). Final FinDer line-source parameters are $L \sim 100$ km, $\Theta = 35°$, and M_{FD} 7.3. Slip distribution in (a) and source time function in (b) and (c) are taken from Zhang, Wang, Chen, *et al.* (2014). The thin black lines show the isoseismals released by the CEA (Wu *et al.*, 2014).

shaking at a given time, so we fill pre-event data gaps with small default noise amplitudes ($0.7\ \mathrm{cm/s^2}$). Figures S1–S3 show the corrected traces along up–down direction of the three earthquakes. We use a total of 378 records for the Wenchuan (distance range: ~20–1800 km), 123 records for the Lushan (~20–800 km), and 66 records for the Jiuzhaigou earthquake (~25–750 km). Stations at large distances (>500 km) only indirectly contribute to FinDer reports by limiting the outer boundary of the area experiencing strong shaking.

Although the spatial coverage with strong-motion sensors is largely uniform and sufficiently dense for the M_w 7.9 Wenchuan and M_w 6.6 Lushan earthquakes (Figs. 2a and 3a), there is a large azimuthal station gap of about 100° west of the epicenter of the M_w 6.5 Jiuzhaigou earthquake (Fig. 4a). Three broadband stations are located in this area at epicentral distances of about 100 km, but their waveforms clipped during the Jiuzhaigou earthquake. We fill this gap by simulating stochastic ground-motion time series (Böse *et al.*, 2012) at the locations of the three broadband station stations, as well as two additional randomly selected sites close to the fault rupture (Fig. 4a, green triangles). In this way, we mimic the possible performance of FinDer in a denser seismic network that reflects what will become available in the near future when the National System for Fast Report of Intensities and EEW project is finished.

To simulate the acceleration time series and thus the temporal evolution of PGA at the five selected sites (Fig. 4a, green triangles), we combine stacked waveform envelopes developed by Cua and Heaton (2009) and Yamada and Heaton (2008) with stochastic simulations, as described in Böse *et al.* (2012). In this approach, the fault rupture is divided into a number of smaller subsources, each radiating *P* and *S* waves when the rupture front arrives. The radiated waves from each subsource are computed from a stochastic time series assuming a simple Brune source model (Brune, 1970), random phase, and a characteristic waveform envelope (Cua and Heaton, 2009). We model the Jiuzhaigou earthquake with three subsources of magnitude *M* 6, corresponding to a subfault length of 10 km (Wells and Coppersmith, 1994), and assume a constant rupture speed of 2.8 km/s. As this is only for testing FinDer, we are not concerned about the details of the rupture or seismic waveforms, but are primarily interested in simulating realistic amplitudes and a representative temporal evolution of PGA. The simulated waveforms are generally in good agreement with strong-motion observations at neighboring stations in terms of amplitude levels and waveform shapes (Fig. S3).

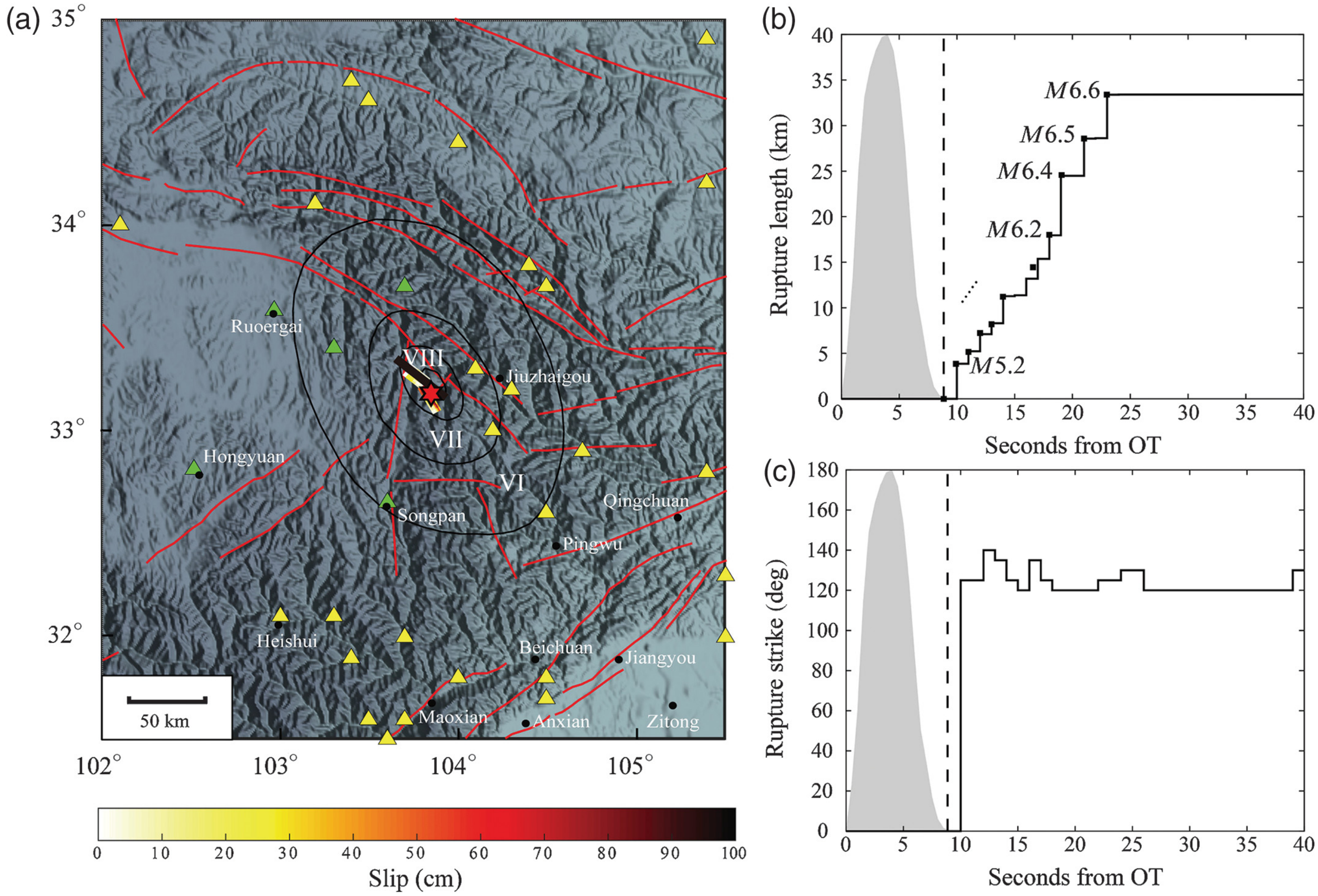


Figure 4. Setting of the 2017 M_w 6.5 Jiuzhaigou earthquake and playback results (follows Fig. 2). Final FinDer line-source parameters are $L \sim 33$ km, $\Theta = 130°$, and M_{FD} 6.6. Green triangles in (a) show stations, where strong-motion data were simulated in this study to constrain FinDer results. Slip distribution in (a) and source time function in (b) and (c) are from Zhang *et al.* (Yong Zhang, written comm., 2018). The thin black lines show the isoseismals released by the CEA (see Data and Resources).

FINDER LINE-SOURCE MODELS

Next, we simulate and analyze the performance of FinDer from waveform playbacks of the Wenchuan, Lushan, and Jiuzahigou earthquakes. We neglect delays caused by data transmission, processing, and alert distribution. In the next section, we discuss the potential application and importance of the resulting line-source models for both EEW and rapid loss estimates. For our tests, we configure FinDer to trigger as soon as ground motions of 20 cm/s^2 are exceeded at two stations (the choice of this threshold is discussed subsequently). The temporal evolution of the FinDer estimated rupture length L and strike Θ as well as the final line source for all three earthquakes are shown in Figures 2–4.

For the 2008 M_w 7.9 Wenchuan earthquake, the stable and final FinDer line-source model (Fig. 2a) is determined ~100 s from simulated OT and has a rupture length of $L \sim 290$ km and a strike of $\Theta = 50°$ (which is equivalent to $\Theta = 50° + 180° = 230°$, because of the symmetry of the line source). Using the empirical relation in equation (2), this rupture length corresponds to a FinDer magnitude of M_{FD} 8.0. The FinDer line-source model is in excellent agreement with the inversion results of Zhang, Wang, Zschau, *et al.* (2014) (M_w 8.0, $L = 300$ km, and $\Theta = 225°$), as well as with the observed aftershock distribution over a length of $L \sim 330$ km in a northeast–southwest direction (Huang *et al.*, 2008). Field surveys following the Wenchuan earthquake identified two separate parallel surface ruptures along the Longmenshan fault zone of L ~240 km and $L \sim 90$ km at N45° E (Xu *et al.*, 2008), which FinDer does not resolve separately. The temporal evolution of FinDer rupture length (and magnitude) and strike (Fig. 2a,b) match well with the rupture duration of ~140 s (Zhang, Wang, Zschau, *et al.*, 2014). See Figure S4 for screenshots of FinDer line-source models at different timesteps.

For the 2013 M_w 6.6 Lushan earthquake, the stable FinDer line-source model (Fig. 3a) is determined within 30 s from OT and has a length of $L \sim 100$ km ($M_{FD} \sim 7.3$) and strike of $\Theta = 35°$ ($= 215°$). Although the strike is in excellent agreement with the value published by the USGS, ($\Theta = 218°$) and the aftershock distribution reported by Fang *et al.* (2013)

($\Theta = \mathrm{N}40°–50°\,\mathrm{E}$), the FinDer rupture length is longer compared to source inversion results of Zhang, Wang, Chen, *et al.* (2014; $L \sim 60$ km), the distribution of relocated aftershocks (Fang *et al.*, 2013) suggesting a length of $L \sim 35$ km, and also compared to a length of the IX contour (∼30 km) in the isoseismal map (Wu *et al.*, 2014), which is one source of estimate of the fault length for earthquakes. Because of the larger acceleration trigger threshold (20 $\mathrm{cm/s^2}$), FinDer triggers shortly after the maximum moment release of the Lushan earthquake (Fig. 3a,b). The total evolution time of ∼25 s of the FinDer rupture is approximately equal to the ∼20 s long rupture duration determined by Zhang, Wang, Chen, *et al.* (2014). See Figure S5 for screenshots.

For the 2017 M_w 6.5 Jiuzhaigou earthquake, stable results are determined within 24 s from OT and are $L \sim 33$ km (M_FD 6.6) and $\Theta = 130°$ (Fig. 4a). These values are in good agreement with the inversion model of Zhang *et al.* (Yong Zhang, written comm., 2018), which suggests rupture along two subparallel faults with strikes of 130° and 150° and a total length of $L \sim 30$ km. Relocated aftershocks show a spread over ∼42 km toward northwest-west direction (Fang *et al.*, 2018). The IX contour in the isoseismal map suggests a ∼25 km long fault rupture (Fig. 4a). Because of both the spatially sparse distribution of stations and the short rupture duration (∼10 s; Yong Zhang, written comm., 2018), FinDer triggers after the rupture terminates (see later discussion on trigger criterion). See Figure S6 for the screenshots of temporal evolution of FinDer line-source models.

DISCUSSION

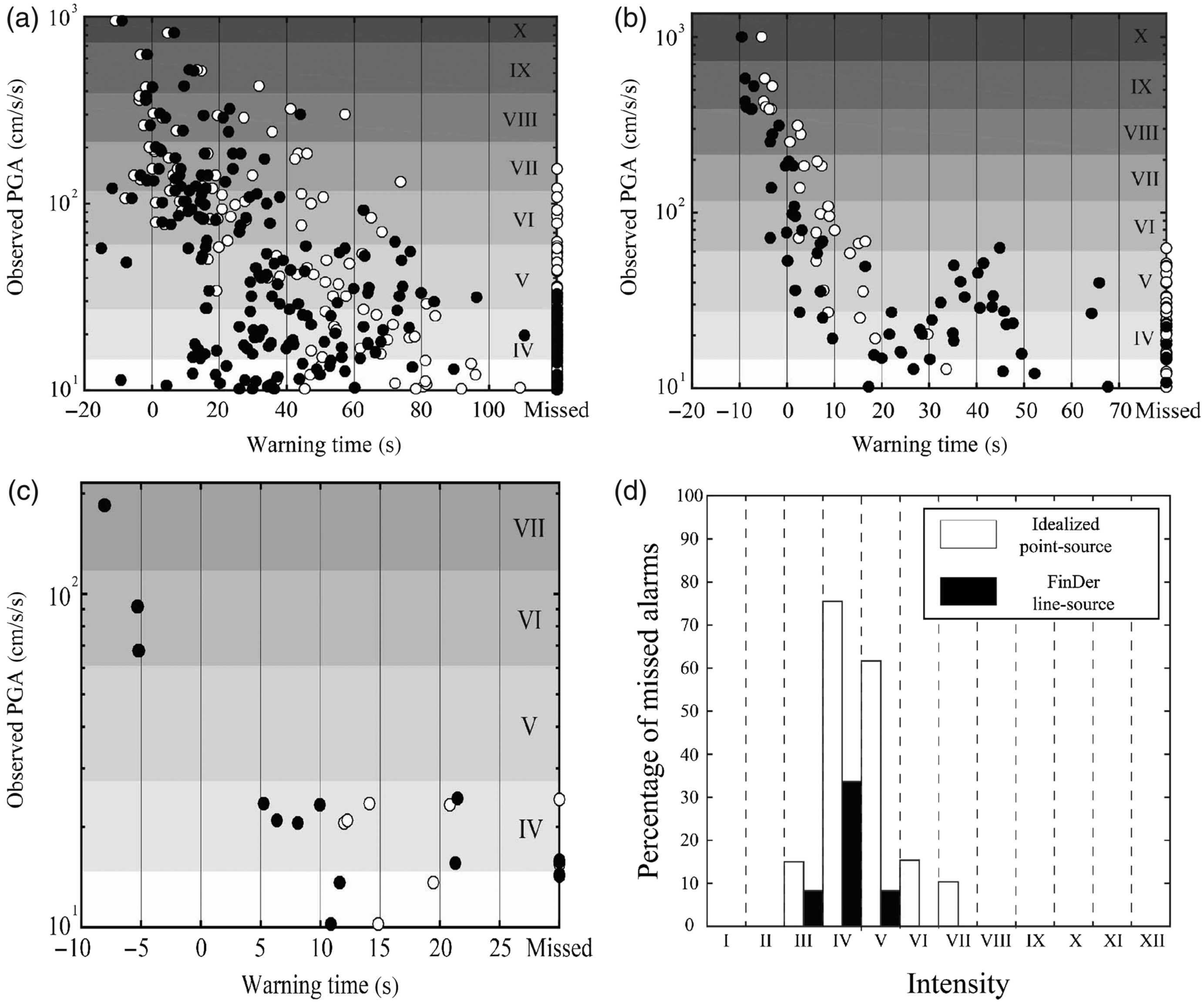
In the following, we discuss the possible benefit of FinDer line-source models for EEW and rapid loss estimates.

FinDer for EEW

The goal of EEW is to issue alerts to areas affected by potentially damaging seismic ground motions before strong shaking initiates. To be effective, warning time must be long enough to take mitigation measures. Sites close to the epicenter usually receive no or very little warning time; they are located in the so-called EEW blind zone (or no-warning zone). With our trigger threshold of 20 $\mathrm{cm/s^2}$, first FinDer results become available at 13, 6, and 10 s from OT, corresponding to blind-zone radii of ∼45 km for the Wenchuan, ∼21 km for the Lushan, and ∼35 km for the Jiuzhaigou earthquake, respectively (assuming an S-wave propagation velocity of $V_S = 3.5$ km/s). Still, as shown in the following, useful warnings can be given to a number of sites outside the blind zone that experience strong shaking.

Figure 5a–c shows the estimated warning time for each of the three earthquakes at each recording station. Although EEW at these stations is not necessarily needed (because there is often no critical settlement or infrastructure nearby), we use the ground-motion time series recorded at these sites for a quantitative assessment of how accurate and timely FinDer ground-motion alerts are. The warning time is defined here as the time interval between the first prediction (which is not necessarily in the first FinDer report) that ground motions will exceed 10 $\mathrm{cm/s^2}$ (MMI II–III), computed from equation (1) with the closest distance to the FinDer line source as a distance metric, and the actual first exceedance of this level. In fact, the good ground-motion prediction performances are obtained at later time than the first prediction, but at the expense of reduced warning time. The soil conditions at the stations were cataloged as soil or rock, and the corresponding regression coefficients in table 3 of Cua and Heaton (2009) are applied to the equation (1) to predict PGA. The occurrence of peak shaking often occurs later, so that this is the minimum warning time the system could provide, assuming zero processing and transmission delays. We select a threshold of 10 $\mathrm{cm/s^2}$, because previous studies have shown that a low threshold generally improves the classification performance for true positives in EEW (Minson *et al.*, 2018). We mainly pick the Cua and Heaton (2009) relation because of its simplicity, but we will see later that this model actually describes the distribution of the observed PGA values in the studied earthquakes surprisingly well. Even though in the long term, it is likely beneficial to build a fully probabilistic EEW system that properly accounts for all sources of uncertainties (if feasible), in the end we will still need to set a simple threshold to decide if or not an alert should be sent out. This decision needs to be done very quickly, because every second counts. The goal of public early warning is to identify potential hazards and risks (a binary 0–1 problem). Knowing the exact level of danger, that is, strength of ground motion, is usually not needed. This may be different in the case of automated responses triggered by EEW, in which different applications and actions depend on different ground-motion levels.

For comparison, we also simulate the performance of an idealized point-source algorithm that requires the P wave to arrive at four stations plus 4 s of waveform data and that produces (unrealistic) errorless estimates of the event location and magnitude once the trigger criterion is fulfilled. Compared to the performance observed in the EEW prototype test system of BCR, this assumption is likely far too optimistic: providing accurate source locations usually requires picks from at least five stations, and reports are typically given after ∼23 s (Peng *et al.*, 2011). Peng, Yang, *et al.* (2017) explored a new τ_c–M_s regression relationship with an expanding P-wave time windows and found that for the Wenchuan earthquake the estimated magnitude reaches a first plateau (M 7.2) approximately 22 s after OT, fluctuates slightly around this value between ∼20 and ∼45 s, and reaches its final estimate of M 7.6 after ∼50 s. The increase and saturation in magnitude suggests that the performance of our idealized point-source algorithm is very optimistic, but it may be used as a rough benchmark to compare to FinDer.



For the point-source algorithm, first estimates would be released after 15 s from OT for the Wenchuan, 9.0 s for the Lushan, and 13 s for the Jiuzhaigou earthquake (Fig. 5, white dots), corresponding to blind-zone radii of ~52, ~32, and ~46 km, respectively. These values are slightly larger for the M_w 6.5 and 6.6 earthquakes compared to the results from the FinDer algorithm (~45 km for the Wenchuan, ~21 km for the Lushan, and ~35 km for the Jiuzhaigou earthquake). The response times of the point-source algorithm during the Wenchuan and Lushan earthquakes are similar to the time when the estimated magnitudes reach M 6.0 and become stable (Peng, Yang, *et al.*, 2017). For the near-source stations, the difference in warning time is small (<5 s). For the three stations with MMI ≥ VI in the Jiuzhaigou earthquake, the difference of available warning times between the two algorithms is extremely small (<1 s), and the circles in Figure 5c overlap.

Compared to FinDer, the idealized point-source algorithm can provide longer warning times for stations with small PGA

Figure 5. Observed peak ground acceleration (PGA) and estimated warning time at each station for the (a) Wenchuan, (b) Lushan, and (c) Jiuzhaigou earthquake based on FinDer line source (black circles) and an idealized point-source algorithm (white circles). The warning time at each station is defined as the time interval between the first prediction (which is not necessarily in the first report) that ground motions will exceed 10 cm/s^2 (corresponding to weak shaking II–III on modified Mercalli intensity [MMI] scale) using ground-motion prediction equations (GMPEs) of Cua and Heaton (2009) and the actual first exceedance of this level. If the observed PGA exceeds that threshold but the system never issues an alert (the predicted PGA does not exceed the threshold), the station is considered to have missed the warning. The occurrence of peak shaking usually occurs later so that this warning time is the minimum useful time the system could provide, assuming zero processing delays. For the point-source algorithm, we assume four stations triggering, 4 s of time delay and perfect estimates of event location and magnitude. (d) Percentage of missed alarms for the two algorithms. The idealized point-source algorithm provides longer warning time than FinDer for many stations with small PGA or low intensity (<VI), but it also produces a much larger number of false and missed alerts (particularly for sites experiencing VI–VII), because it neglects source dimensions.

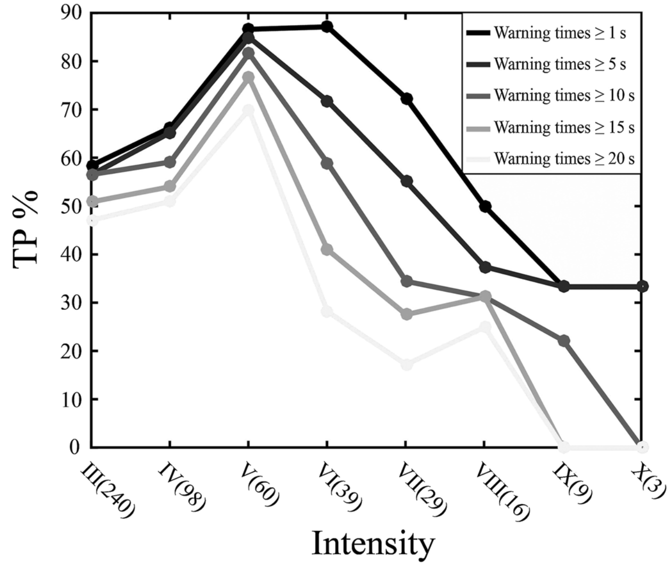


Figure 6. Percentage of true positives (TP) for different levels of intensity (MMI) and different ranges of warning time using FinDer models for the Wenchuan, Lushan, and Jiuzhaigou earthquakes. Numbers in brackets on *x* axis give counts of stations for each MMI bin. These stations, which are considered as TP, do not include stations that release missed alerts, false alerts (observed PGA ≤ 10 cm/s^2, but predicted PGA ≥ 10 cm/s^2), and with no need for warnings (both of observed and predicted PGA ≤ 10 cm/s^2). If the available set of strong-motion data had been available in real time and without delay, 50%–80% of sites experiencing shaking intensities of IV–VII could receive ≥10 s of warning, and 30% of sites experiencing VII–IX could receive ≥5 s. Warning time is expected to increase once the full seismic network is deployed.

or low intensity (<VI), but it also produces a much larger number of false and missed alerts, particularly for sites experiencing damaging shaking, mainly because source dimensions are neglected (Fig. 5d). For all three earthquakes, FinDer misses 12% of true positives, the point-source algorithm 31% (including many with MMI ≥ V; Fig. 5d). Once a denser station network is deployed, the FinDer trigger threshold (20 cm/s^2 at two neighboring stations) should be reduced and warning times are expected to increase.

Figure 6 shows the ratio of early warning stations for different ranges of warning time (≥1, ≥5, ≥10, ≥15, and ≥20 s) to the total number of stations per MMI bin (true positives) for all three earthquakes. Although the reported intensities are measured on the Chinese intensity scale, the difference to MMI is small (Hu, 2009), and we therefore assume that they are approximately equal. If the processed set of strong-motion data had been available in real time, more than 50%–80% of sites experiencing shaking intensities of IV–VII could receive ≥10 s of warning, and 30% of sites experiencing VII–IX could receive ≥5 s of warning (Fig. 6). Some stations located in high-intensity areas (IX–X), distant from the epicenter but close to the rupturing fault (or asperities), could obtain an effective warning time of ~1–10 s. If the gray (soft) blind zone, which takes the potential failure of seismic stations into account, was properly controlled (Li and Wu, 2016; Li, Zhang, and Zhang, 2018), the disaster risk in these high-intensity areas could be significantly reduced by an EEW system.

The warning time shown in Figures 5 and 6 could be increased by several seconds, if a smaller trigger threshold was chosen. The majority of FinDer installations around the world use a much smaller threshold of 2 cm/s^2. However, the current station density around the Wenchuan earthquake is quite sparse, and for several seconds only a single seismic station (WCW) reports very large amplitudes (PGA = 957 cm/s^2), whereas ground motions at neighboring stations within ~25 km are weak, because the *S* wave has not yet arrived. In fact, Figure S1 shows that the nearest station to WCW with large PGA is station SFB, which is about 50 km away from WCW (Fig. 2a). As a consequence, if a smaller threshold of, for instance, 2 cm/s^2 was selected, FinDer would flag this critical station as noisy and discard it from computations. This problem can be avoided by selecting a larger trigger threshold as is done here (20 cm/s^2), which, however, is at the cost of delaying the FinDer trigger and reducing warning time. Once more stations are deployed in Sichuan province in the near future, a lower trigger threshold will be preferred.

Unilateral fault ruptures like in the Wenchuan earthquake, in which the rupture size is larger than the blind zone, offer the best opportunity for EEW, because people and users (for example high-speed railways) in some meizoseismal regions (that is, areas of maximum damage), which are close to the fault rupture but distant from the epicenter, could receive effective warnings (Fig. 6). In this case, if appropriate preventive actions can be taken in time, the number of casualties and infrastructure damage could be reduced. Warning time for sites within the rupture zone is discussed in Ma *et al.* (2011). Taking the seismic extended source dimensions and the potential failure of seismic stations into account, Li and Wu (2016) and Li, Zhang, and Zhang (2018) proposed the existence of a gray (soft) blind zone surrounding the theoretical black (hard) blind zone that can be controlled and minimized by increasing the seismic station density.

FinDer estimates the current dimensions and orientation of an ongoing earthquake rupture from near-source observed ground motions. This means that warning time results mainly from the time difference between the onset of near-source observed ground motions, which are used to constrain the FinDer line source, and the onset of strong motions at larger distances. The current version of FinDer does not predict the future evolution of this rupture, which would be beneficial, because warning time could then be increased and areas affected by the earthquake could be better constrained. Böse and Heaton (2010) and Hutchison *et al.* (2019) have started to investigate the potential of probabilistic rupture prediction by incorporating fault properties, such as fault maturity, into FinDer, but this approach needs further development and testing.

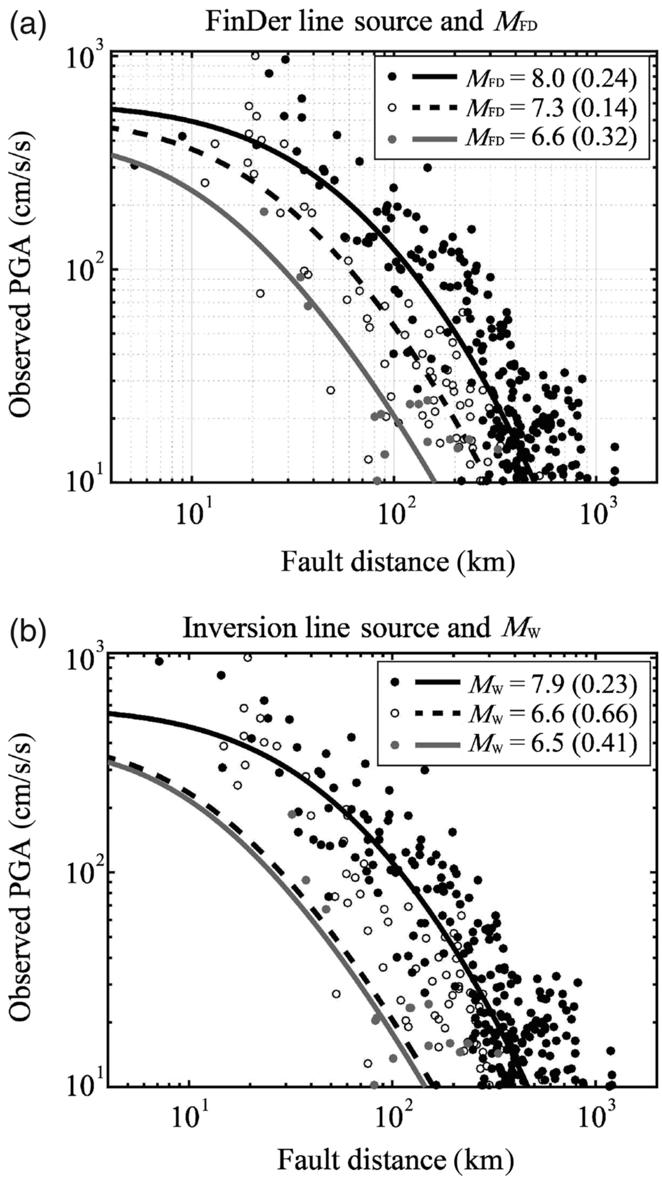


Figure 7. Observed PGA for the M_w 7.9 Wenchuan (black dots), 2013 M_w 6.6 Lushan (white dots), and 2017 M_w 6.5 Jiuzhaigou (gray dots) earthquakes as a function of fault distance from (a) FinDer line source and (b) inversion line source (Zhang, Wang, Chen, *et al.*, 2014; Zhang, Wang, Zschau, *et al.*, 2014; Y. Zhang, written comm., 2018). Lines show theoretical values modeled from GMPEs after Cua and Heaton (2009) using M_{FD} and M_w, respectively. The root mean square (rms) errors (see legend) are much smaller for the FinDer model compared to the results from the event catalog (in particular for the Lushan earthquake), suggesting that the FinDer model is well suited to explain the observed spatial distribution of high-frequency motions, which are of engineering concern and are thus critical to earthquake early warning (EEW), but may in some earthquakes disagree with M_w. Although the use of a regional GMPE is generally preferred, the fit with the Cua and Heaton (2009) model is surprisingly good.

FinDer magnitude (M_{FD})

For all three earthquakes analyzed in this article, the FinDer magnitude M_{FD} is somewhat larger than the reported moment magnitude M_w (Wenchuan: M_{FD} 8.0, M_w 7.9; Lushan: M_{FD} 7.3, M_w 6.6; and Jiuzhaigou: M_{FD} 6.6, M_w 6.5). However, for the Wenchuan and Jiuzhaigou earthquakes, the difference between M_{FD} and M_w is negligible, because M_w is typically determined with an accuracy of 0.2 units. For the Lushan earthquake, however, M_{FD} exceeds M_w beyond the usual variations of magnitude estimates. It is important, however, that the FinDer line-source models are constructed to characterize the distribution of high-frequency (PGA) motions (Fig. S7) rather than to provide an accurate source characterization (Böse *et al.*, 2018). These high-frequency motions are of engineering concern and as such are critically important to EEW. Because of its high-frequency physical background, M_{FD} is in better agreement with the energy magnitude (Picozzi *et al.*, 2017) rather than with the long-period M_w. In general, M_{FD} should be understood as a scaling factor that quantifies the level and spatial distribution of high-frequency ground motions (Böse *et al.*, 2018). We do not necessarily expect close agreement with other magnitude scales, especially with those that are sensitive to longer periods, including M_w.

For a better illustration of this statement, we plot in Figure 7 the observed PGA values for all three earthquakes as a function of fault distance. We compare these amplitudes with theoretical values, which we model from GMPEs after Cua and Heaton (2009) (which we used for FinDer template generation) using M_{FD} or M_w, respectively. The root mean square (rms) errors for the observed and modeled logarithmic amplitudes are much smaller for the higher M_{FD} (Wenchuan: 0.24, Lushan: 0.14, and Jiuzhaigou: 0.32) compared to the lower M_w (Wenchuan: 0.23, Lushan: 0.66, and Jiuzhaigou: 0.41). The difference is most obvious for the M_w 6.6/M_{FD} 7.3 Lushan earthquake, in which PGA is significantly underestimated for the smaller M_w. In fact, the intensity map of the Lushan earthquake released by CEA (isoseismals in Fig. 3) also suggests an elongated pattern of high intensities toward the southwest from the epicenter, which likely resulted from a combined effect of local site conditions, hanging wall, topography (Wen and Ren, 2014), as well as rupture directivity. The high-PGA values in the southwestern direction are largely responsible for the high M_{FD} in the Lushan earthquake.

We compared the rupture length–magnitude relationships of different earthquake type datasets in Wells and Coppersmith (1994). The magnitude difference obtained by selecting different regression parameters is small, basically 0.1–0.2 magnitude units. We believe that this difference is negligible and that the specific source mechanism information will not have much impact on our current results.

FinDer for rapid loss estimates

Aside from EEW, FinDer line-source models could also provide an important contribution to rapid loss and casualty estimates following great earthquakes like in Wenchuan. This information is critical to decision makers, responders, and rescue teams to ensure rapid mobilization commensurate with the

TABLE 2
QLARM-Estimated and Reported Casualties for the 2008 M_w 7.9 Wenchuan Earthquake Using Different Source Models

Source Model	Fatalities	Injured	Sources
Point source	4,000–20,000	12,000–61,000	CENC, USGS
Line source from inversion	33,000–100,000	88,000–267,000	Zhang, Wang, Zschau, *et al.* (2014)
Line source from FinDer	29,000–86,000	77,000–232,000	–
Reported	87,149	374,643	Chen and Booth (2011)

See the FinDer for rapid loss estimates section for explanation. QLARM, Quake Loss Assessment for Response and Mitigation.

TABLE 3
PAGER-Estimated and Reported Casualties and Economic Losses for the 2008 M_w 7.9 Wenchuan Earthquake Using Different Source Models

Source Model		0–1	1–10	10–10^2	10^2–10^3	10^3–10^4	10^4–10^5	10^5–10^6	Sources
Point source	Fatalities	–	–	–	9%	31%	38%	21%	CENC, USGS
	Economic losses	–	–	–	–	10%	27%	61%	
Line source from inversion	Fatalities	–	–	–	3%	18%	38%	40%	Zhang, Wang, Zschau, *et al.* (2014)
	Economic losses	–	–	–	–	8%	25%	66%	
Line source from FinDer	Fatalities	–	–	–	10%	32%	37%	19%	–
	Economic losses	–	–	–	–	10%	28%	60%	–
Reported	Fatalities	87,149							Chen and Booth (2011)
	Economic losses	~1.5×10^6 million U.S.							

Economic losses are millions in U.S. dollars. All estimates are given in probabilities. See the FinDer for rapid loss estimates section for explanation. PAGER, Prompt Assessment of Global Earthquakes for Response.

extent of the disaster (Wyss, 2014). Initial loss alerts from both Quake Loss Assessment for Response and Mitigation (QLARM; Wyss *et al.*, 2006; Wyss and Zibzibadze, 2009; Trendafiloski *et al.*, 2011; Wyss, 2014) and Prompt Assessment of Global Earthquakes for Response (PAGER; Wald *et al.*, 2008; Jaiswal and Wald, 2010, 2011; Wald, 2010) are typically released within about 30 min from OT and are based on an earthquake point-source approximation. These initial estimates are usually accurate enough for smaller earthquakes ($L < \sim 40$ km, $M < 7$), but are prone to underestimation in larger magnitude events due to the negligence of the extended radiation of energy along the length of the fault rupture. It usually takes a few hours to days before extended source models are constrained from either early aftershocks or waveform inversion, and loss estimates are updated. Automatic procedures might be able to reduce these delays.

For the 2008 M_w 7.9 Wenchuan earthquake, QLARM released a first casualty estimate with 40,000–90,000 fatalities and 97,000–214,000 injured 100 min from OT based on a point source and low-attenuation model. PAGER, which in 2008 was not yet reporting fatality estimates, distributed an alert 30 min after OT stating that 1.2 million people were exposed to MMI VIII or greater; half a day later, based on a simple finite-fault model derived from early aftershocks and waveform inversion, PAGER updated this estimate to 5.2 million people, and predicted fatalities in the tens of thousands (Earle *et al.*, 2008). As of 25 August 2008, the State Council of China reported 87,149 people dead (including 17,923 missing) and at least 374,643 injured. The direct economic loss was estimated as $1.5 trillion U.S. (Tables 2 and 3; Chen and Booth, 2011).

Using the example of the M_w 7.9 Wenchuan earthquake, we adopt the QLARM and PAGER software to retrospectively predict earthquakes losses and to compare them with the official reports. We use three earthquake source models: (1) a point-source approximation, (2) an extended fault model derived from waveform inversion, and (3) the FinDer line-source model. For the point-source model, we use the epicenter and M_w published by CENC and USGS, respectively. The inversion line-source model is placed along the surface break and approximately within the center of the simplified source rectangle of Zhang, Wang, Zschau, *et al.* (2014). For FinDer, we use the final line-source model as shown in Figure 2. All model parameters are given in Table S1.

Loss estimates from QLARM are summarized in Table 2. Averaging between different attenuation characteristics on the mountain versus plain sides along the fault rupture, we set the QLARM attenuation to $C_3 = 3.7$ (Trendafiloski *et al.*, 2011; Wyss, 2014 for details). We also use corrected building and population distribution models as described in Li, Wyss *et al.* (2018), which explains why the loss estimates for the point-source model in Table 2 are different from the real-time estimates previously described. The ranges of the estimated fatalities and injured are the formal limits for the result being within 95% probability, which is typically adopted by QLARM to account for the uncertainties in the source and ground-motion characteristics (Wyss *et al.*, 2011; Wyss, 2014). The loss estimates based on the point-source model are underestimated by a factor of more than four in fatalities and of six in injuries (Table 2). By contrast, the loss estimates based on the FinDer line-source model are similar to those based on the inversion line source.

Loss estimates from PAGER are summarized in Table 3, and the corresponding one-PAGER reports are shown in Figure S8. The fault width would likely not be known in real time, so based on the aftershock distribution and inversion rupture model, we use the approximation value of 20 and 30 km for the source model (2) and (3), respectively, in ShakeMap, the shaking input to PAGER. For all three source models, loss estimates reach the highest PAGER alert level (red alert). Using the point-source model, PAGER predicts casualties on the order of 1000–10,000 with an occurrence probability of 31%, and 10,000–100,000 casualties with a probability of 38%; economic losses are estimated as $\$10^5$–$10^6$ million U.S. Using the inversion model (10,000–100,000 casualties: 38% probability; 100,000–1,000,000 casualties: 40% probability), the number of fatalities and the estimated economic losses are in the same range as those reported. For the FinDer model, the estimates of both the fatalities (1000–10,000, 32% probability; 10,000–100,000, 37% probability) and the economic losses ($\$10^5$–$10^6$ million U.S. with 60% probability) are consistent with the reports ($\$1.5 \times 10^6$ million U.S.).

FinDer line-source models could be calculated within the order of a few minutes (e.g., ~2 min from OT for the Wenchuan earthquake), much faster than the inversion models (hours to days from OT; the same is true for models based on aftershock distributions), and as such could provide an important improvement to the existing early loss estimates in the aftermath of devastating earthquakes.

For ShakeMap (and thus PAGER) purposes specifically, dimension of earthquake rupture is most useful to add when the map is dominated by predictions rather than observations. When there are sufficient real-time stations, ShakeMap, which is a combination of predictions and observations, dictates the shaking pattern rather than the estimates, so getting the predictions to be sounder is always helpful. However, there are always areas where there are few stations and local or regional strong-motion data are not available, and, thus, where estimates are important.

CONCLUSIONS

In this study, we ran playbacks of local and regional onscale strong-motion waveforms recorded by CSMNC during the M_w 7.9 Wenchuan, M_w 6.6 Lushan, and M_w 6.5 Jiuzhaigou earthquakes to study the performance of the FinDer algorithm (Böse *et al.*, 2012, 2015, 2018). We find that the FinDer line-source models are generally in good agreement with the observed spatial distribution of aftershocks and finite-fault models determined from waveform inversion (Figs. 2–4), and as such are well suited as early estimates of extended fault ruptures, in particular, in ruptures exceeding 50 km in length. The FinDer models provide useful information for both EEW and rapid loss estimates. We attribute the overestimation of the rupture size of the Lushan earthquake to the unusually high radiation of high-frequency motions in this event. By design, FinDer determines a line-source model that is best suited to explain high-frequency motions.

EEW for large earthquakes is challenging, because a proper characterization of the extended source is needed to predict ground motions. However, compared to moderate earthquakes, many more users and the public could benefit from EEW in large earthquakes like in Wenchuan and warning time might be long enough to respond (e.g., Heaton, 1985). If the set of strong-motion data for the three earthquakes had been available in real time, more than 50%–80% of sites experiencing shaking intensities of MMI IV–VII could have received ≥10 s of warning, and 30% of sites experiencing MMI VII–IX could have received ≥5 s of warning. Even some high-intensity areas (MMI IX and X), distant from the epicenter but close to the fault rupture (or asperities), could receive effective warning time of ~1–10 s. We expect that both warning time and the accuracy of source and ground-motion estimates will be improved, once the seismic network as planned within the National System for Fast Report of Intensities and EEW project is fully deployed.

Aside from EEW, FinDer models can also help provide quick and accurate loss estimates for rapid response in the aftermath of devastating earthquakes. We compared the loss estimates for the M_w 7.9/M_s 8.0 Wenchuan earthquake using a simple point-source model (as used in the initial loss estimates in QLARM and PAGER), an inversion line-source model, and the FinDer line-source model. QLARM results show that the loss estimates based on the point-source model are more than four times underestimated in fatality and more than six times underestimated in injuries, whereas estimates based on the FinDer line source are as accurate as those based on inversion models. Although inversion models are typically generated hours to days after the earthquake hit, FinDer line-source models could be computed and used in loss estimate software within a few minutes from the event origin, if a local seismic real-time network existed.

Although EEW needs to be issued within a few seconds from the event origin, earthquake loss estimates can be provided in a few minutes. This additional time window for loss estimates provides the opportunity to verify the compliance of the FinDer estimated rupture length with magnitude estimates from other agencies. Estimating azimuth and end points of a fault almost instantly offers a significant advantage for calculating expected losses. For rupture length >50 km, this is obvious because the nearest distance to the energy release along the rupture can then be calculated for each settlement in minutes when the earthquake occurs in the middle of a populated area. Even for ruptures shorter than 50 km, knowledge of the endpoints can lead to an order of magnitude improvement in estimating fatalities. During the last 15 yr, it has happened repeatedly that epicenters of M 6.5+ earthquakes were reported offshore, in a desert or in uninhabited mountains. With a point source as the basis of the loss estimates, the number of fatalities was expected to be small or zero, but the rupture ran up to a dense population center, resulting in more than 1000 deaths, for example, the Yogyakarta earthquake of M_w 6.4 on 26 May 2006. Epicenter uncertainties can map into more than 10,000 fatalities underestimated (Wyss *et al.*, 2006, 2011, 2012). With endpoints of the rupture determined by FinDer, these errors can be avoided, provided that seismic waveform data are available in real time.

With the increasing number of countries building dense networks of strong-motion stations with low latency, FinDer has the potential to significantly improve both traditional EEW and rapid loss estimates, if operated in these networks.

DATA AND RESOURCES

Strong-motion waveforms used in this study were provided by the China Strong Motion Network Center (CSMNC) at the Institute of Engineering Mechanics (China Earthquake Administration [CEA]). Information on seismic broadband stations was obtained from the Data Management Centre of China National Seismic Network at Institute of Geophysics (IGP, SEISDMC, doi: 10.11998/SeisDmc/SN) at CEA. Background terrain in Figures 1–4 uses the gtopo30 data from the USGS Earth Explorer (https://earthexplorer.usgs.gov, last accessed January 2019). The isoseismals in Figure 4 are taken from CEA (http://www.chinanews.com/gn/2017/08-12/8303257.shtml, last accessed: March 2019). See supplemental material for plots of strong-motion records of the three earthquakes studied, the temporal evolution of finite-fault rupture detector (FinDer) line-source models, source parameters used for the loss estimates of the Wenchuan earthquake as well as Prompt Assessment of Global Earthquakes for Response (PAGER) results.

ACKNOWLEDGMENTS

The authors would like to thank Yong Zhang (Peking University) for providing us with the slip distributions and source time functions of the Wenchuan, Lushan, and Jiuzhaigou earthquakes, Chaoyong Peng (Institute of Geophysics [IGP], China Earthquake Administration [CEA]) for discussions about the technical details of the National System for Fast Report of Intensities and Earthquake Early Warning (EEW) project, Liangqiong Lou (China Strong Motion Networks Center [CSMNC]) for information about the operations of strong-motion networks, and Yan Zhang (IGP, CEA) for helping us to apply for the data. The authors would also like to thank Stefan Wiemer, Alireza Khodaverdian, Frédérick Massin (Swiss Federal Institute of Technology [ETH] Zürich), Thomas H. Heaton (Caltech), Gavin P. Hayes, and Janet L. Slate (U.S. Geological Survey [USGS]) for related scientific discussions, as well as Associate Editor Stefano Parolai and four anonymous reviewers for their feedback and suggestions. Jiawei Li also thanks the China Scholarship Council (201804190032) and the Swiss Seismological Service (SED) for funding his visit at ETH Zürich.

REFERENCES

Allen, R. M. (2007). The ElarmS earthquake early warning methodology and application across California, in *Earthquake Early Warning Systems*, P. Gasparini, G. Manfredi, and J. Zschau (Editors), Springer, Berlin, Germany, New York, New York, 21–43.

Allen, R. M., and D. Melgar (2019). Earthquake early warning: Advances, scientific challenges, and societal needs, *Annu. Rev. Earth Planet. Sci.* **47,** 361–388.

Allen, R. M., P. Gasparini, O. Kamigaichi, and M. Böse (2009). The status of earthquake early warning around the world: An introductory overview, *Seismol. Res. Lett.* **80,** 682–693.

Behr, Y., J. F. Clinton, P. Kästli, C. Cauzzi, R. Racine, and M.-A. Meier (2015). Anatomy of an earthquake early warning (EEW) alert: Predicting time delays for an end-to-end EEW system, *Seismol. Res. Lett.* **86,** 830–840.

Böse, M., and T. H. Heaton (2010). Probabilistic prediction of rupture length, slip and seismic ground motions for an ongoing rupture: Implications for early warning for large earthquakes, *Geophys. J. Int.* **183,** 1014–1030.

Böse, M., C. Felizardo, and T. H. Heaton (2015). Finite-fault rupture detector (FinDer): Going real-time in Californian ShakeAlert warning system, *Seismol. Res. Lett.* **86,** 1692–1704, doi: 10.1785/0220150154.

Böse, M., T. H. Heaton, and E. Hauksson (2012). Real-time finite fault rupture detector (FinDer) for large earthquakes, *Geophys. J. Int.* **191,** 803–812, doi: 10.1111/j.1365-246X.2012.05657.x.

Böse, M., C. Ionescu, and F. Wenzel (2007). Earthquake early warning for Bucharest, Romania: Novel and revised scaling relations, *Geophys. Res. Lett.* **34,** L07302, doi: 10.1029/2007GL029396.

Böse, M., D. E. Smith, C. Felizardo, M.-A. Meier, T. H. Heaton, and J. F. Clinton (2018). FinDer v.2: Improved real-time ground-motion predictions for M2-M9 with seismic finite-source characterization, *Geophys. J. Int.* **212,** 725–742, doi: 10.1093/gji/ggx430.

Brune, J. N. (1970). Tectonic stress and the spectra of seismic shear waves from earthquakes, *J. Geophys. Res.* **75,** 4997–5009.

Chen, Y., and D. C. Booth (2011). *The Wenchuan earthquake of 2008: Anatomy of a disaster*, Science Press cooperating with Springer, Beijing, China.

Chung, A. I., I. Henson, and R. M. Allen (2019). Optimizing earthquake early warning performance: ElarmS-3, *Seismol. Res. Lett.* **90,** 727–743, doi: 10.1785/0220180192.

Clinton, J., A. Zollo, A. Marmureanu, C. Zulfikar, and S. Parolai (2016). State-of-the art and future of earthquake early warning in the European region, *Bull. Earthq. Eng.* **14,** 2441–2458.

Cua, G. B., and T. H. Heaton (2009). *Characterizing Average Properties of Southern California Ground Motion Amplitudes and Envelopes*, Earthquake Engineering Research Laboratory, Pasadena, California, available at http://resolver.caltech.edu/CaltechEERL:EERL-2009-05 (last accessed February 2019).

Deng, Q. D., P. Z. Zhang, Y. K. Ran, X. P. Yang, W. Min, and L. C. Chen (2003). Active tectonics and earthquake activities in China, *Earth Science Frontiers* **10,** 66–73 (in Chinese with English abstract).

Department of Earthquake Disaster Prevention of the State Seismological Bureau (1995). *The Catalogue of Chinese Historical Strong Earthquakes*, Seismological Press, Beijing, China (in Chinese with English abstract).

Department of Earthquake Disaster Prevention of the China Earthquake administration (1999). *The Catalogue of Chinese Modern Earthquakes* (in Chinese with English abstract), China Science and Technology Press, Beijing, China.

Earle, P. S., D. J. Wald, T. L. Allen, K. S. Jaiswal, K. A. Porter, and M. G. Hearne (2008). Rapid exposure and loss estimates for the May 12, 2008 M_w 7.9 Wenchuan earthquake provided by the U.S. Geological Survey's PAGER system, *The 14th World Conf. on Earthquake Engineering*, Beijing, China, 12–17 October 2008, available at http://www.iitk.ac.in/nicee/wcee/article/14_S31-039.pdf (last accessed January 2019).

Erdik, M., Y. Fahjan, O. Ozel, H. Alcik, A. Mert, and M. Gul (2003). Istanbul earthquake rapid response and the early warning system, *Bull. Earthq. Eng.* **1,** 157–163, doi: 10.1023/A:1024813612271.

Espinosa-Aranda, J. M., A. Jimenez, G. Ibarrola, F. Alcantar, A. Aguilar, M. Inostroza, and S. Maldonado (1995). Mexico City seismic alert system, *Seismol. Res. Lett.* **66,** 42–53.

Fang, L. H., J. P. Wu, J. R. Su, M. M. Wang, C. Jiang, L. P. Fan, W. L. Wang, C. Z. Wang, and L. X. Tan (2018). Relocation of mainshock and aftershock sequence of the MS7.0 Sichuan Jiuzhaigou earthquake, *Chin. Sci. Bull.* **63,** 649–662 (in Chinese with English abstract).

Fang, L. H., J. P. Wu, W. L. Wang, Z. Y. Lü, C. Z. Wang, T. Yang, and Y. Cai (2013). Relocation of the mainshock and aftershock sequences of MS7.0 Sichuan Lushan earthquake, *Chin. Sci. Bull.* **58,** 3451–3459.

Fang, R. X., C. Shi, G. X. Wang, and J. N. Liu (2014). Epicenter and magnitude of large earthquake determined from high-rate GPS observations: A case study of the 2008 M 8.0 Wenchuan earthquake, *Sci. China Earth Sci.* **57,** 1645–1652.

Gao, M. T., G. C. Wu, X. Y. Wu, and C. Lang (2014). From the perspective of social risk theory: The role of the earthquake prevention and disaster reduction system in the national significant seismic monitoring and protection regions, *Earthq. Res. China* **30,** 300–305 (in Chinese with English abstract).

Given, D., R. M. Allen, A. S. Baltay, P. Bodin, E. S. Cochran, K. Creager, R. M. de Groot, L. S. Gee, E. Hauksson, T. H. Heaton, *et al.* (2018). Implementation plan for the ShakeAlert system-An earthquake early warning system for the West Coast of the United States, *U.S. Geol. Surv. Open-File Rept. 2018-1155*, available at https://pubs.usgs.gov/of/2018/1155/ofr20181155.pdf (last accessed January 2019).

Han, L. B., J. Cheng, Y. R. An, L. H. Fang, C. S. Jiang, B. Chen, Z. L. Wu, J. Liu, X. W. Xu, R. F. Liu, *et al.* (2018). Preliminary report on the 8 August 2017 MS 7.0 Jiuzhaigou, Sichuan, China, earthquake, *Seismol. Res. Lett.* **89,** 557–569.

Heaton, T. H. (1985). A model for a seismic computerized alert network, *Science* **228,** 987–990, doi: 10.1126/science.228.4702.987.

Horiuchi, S., H. Negishi, K. Abe, A. Kamimura, and Y. Fujinawa (2005). An automatic processing system for broadcasting earthquake alarms, *Bull. Seismol. Soc. Am.* **95,** 708–718, doi: 10.1785/0120030133.

Hoshiba, M., and S. Aoki (2015). Numerical shake prediction for earthquake early warning: Data assimilation, real-time shake-mapping, and simulation of wave propagation, *Bull. Seismol. Soc. Am.* **105,** 1324–1338, doi: 10.1785/0120140280.

Hu, Y. X. (2009). *Earthquake Engineering*, Second Ed., Seismological Press, Beijing, China (in Chinese).

Huang, Y., J. P. Wu, T. Z. Zhang, and D. N. Zhang (2008). Relocation of M 8.0 Wenchuan earthquake and its aftershock sequence, *Sci. China Earth Sci.* **51,** 1703–1711.

Hutchison, A., M. Böse, J. W. Li, and J. F. Clinton (2019). Real-time slip profiles from back-projection of dynamic displacement amplitudes derived from strong-motion waveforms: Demonstration for the 2008 MW 7.9 Wenchuan (China) earthquake, *2019 EGU General Assembly*, Vienna, Austria, 7–12 April 2019, EGU2019–16273.

Ionescu, C., M. Böse, F. Wenzel, A. Marmureanu, A. Grigore, and G. Marmureanu (2007). An early warning system for deep Vrancea (Romania) earthquakes, in *Earthquake Early Warning Systems*, P. Gasparini, G. Manfredi, and J. Zschau (Editors), Springer, Berlin, Germany, New York, New York, 342–349.

Jaiswal, K. S., and D. J. Wald (2010). An empirical model for global earthquake fatality estimation, *Earthq. Spectra* **26,** 1017–1037.

Jaiswal, K. S., and D. J. Wald (2011). Rapid estimation of the economic consequences of global earthquakes, *U.S. Geol. Surv. Open-File Rept. 2011-1116*, available at https://pubs.usgs.gov/of/2011/1116/downloads/OF11-1116_508.pdf.

Kamigaichi, O., M. Saito, K. Doi, T. Matsumori, and S. Tsukada (2009). Earthquake early warning in Japan: Warning the general public and future prospects, *Seismol. Res. Lett.* **80,** 717–726.

Leyton, F., S. Ruiz, J. C. Baez, G. Meneses, and R. Madariaga (2018). How fast can we reliably estimate the magnitude of subduction earthquakes?, *Geophys. Res. Lett.* **45,** 9633–9641, doi: 10.1029/2018GL078991.

Li, H. J., J. J. Zhang, and Y. L. Tang (2017). Testing earthquake early warning parameters, $\tau_p^{\max}$, τ_c, and P_d, for rapid magnitude estimation in the Sichuan, China, region, *Bull. Seismol. Soc. Am.* **107,** 1439–1450.

Li, J. W., and Z. L. Wu (2016). Controlling the 'blind zone' of an earthquake early warning system (EEWS): A case study of the Beijing Capital Circle prototype EEWS (in Chinese with English abstract), *Earthq. Res. China* **32,** 584–594.

Li, J. W., M. Wyss, Z. L. Wu, and S. Y. Zhou (2018). Estimated casualty risk for six scenario great earthquakes in Sichuan–Yunnan region, China, for disaster preparation, *International Conf. for the Decade Memory of the Wenchuan Earthquake with The 4th International Conf. on Continental Earthquakes (The 4th ICCE) and The 12th General Assembly of the Asian Seismological Commission (ASC)*, Chengdu, Sichuan, China, 12–14 May 2018, the 4th ICCE–0000000094.

Li, J. W., S. F. Zhang, and Y. Zhang (2018). Seismic finite source and its significance to earthquake early warning system (EEWS), *Acta Seismol. Sinica* **40,** 728–736, doi: 10.11939/jass.20180010 (in Chinese with English abstract).

Li, X. J., Z. H. Zhou, M. Huang, R. Z. Wen, H. Y. Yu, D. W. Lu, Y. N. Zhou, and J. W. Cui (2008). Preliminary analysis of strong-motion recordings from the magnitude 8.0 Wenchuan, China, earthquake of 12 May 2008, *Seismol. Res. Lett.* **79,** 844–854, doi: 10.1785/gssrl.79.6.844.

Ma, X. J., Z. L. Wu, H. S. Peng, and T. F. Ma (2011). Challenging the limit of EEW: A scenario of EEWS application based on the lessons of the 2008 Wenchuan earthquake, in *Advances in Geosciences*, C. H. Lo (Editor), Vol. 31, Solid Earth Science, World Scientific Publishing, Hackensack, New Jersey, 11–22.

Ma, Z. J., and A. X. Zhao (1991). The general situation of earthquake disasters in China and suggestions for the countermeasures of disaster mitigation, *Earthq. Res. China* **7,** 89–94 (in Chinese with English abstract).

Massin, F., J. F. Clinton, J. Porras, G. Marroquin, W. Strauch, and M. Böse (2019). Evaluating and improving earthquake early warning in Central America, *2019 SSA Annual Meeting*, Seattle, Washington, 23–26 April.

Minson, S. E., M.-A. Meier, A. S. Baltay, T. C. Hanks, and E. S. Cochran (2018). The limits of earthquake early warning: Timeliness of ground motion estimates, *Sci. Adv.* **4,** eaaq0504, doi: 10.1126/sciadv.aaq0504.

Nazeri, S., S. Colombelli, and A. Zollo (2019). Fast and accurate determination of earthquake moment, rupture length and stress release for the 2016–2017 Central Italy seismic sequence, *Geophys. J. Int.* **217,** 1425–1432.

Peng, C. Y., Y. Chen, Q. S. Chen, J. S. Yang, H. T. Wang, X. Y. Zhu, Z. Q. Xu, and Y. Zheng (2017). A new type of tri-axial accelerometers with high dynamic range MEMS for earthquake early warning, *Comput. Geosci.* **100,** 179–187.

Peng, C. Y., J. S. Yang, B. Xue, X. Y. Zhu, and Y. Chen (2014). Exploring the feasibility of earthquake early warning using records of the 2008 Wenchuan earthquake and its aftershocks, *Soil Dynam. Earthq. Eng.* **57,** 86–93.

Peng, C. Y., J. S. Yang, Y. Zheng, X. Y. Zhu, and Y. Chen (2017). New τ_c regression relationship derived from all P-wave time windows for rapid magnitude estimation, *Geophys. Res. Lett.* **44,** 1724–1731.

Peng, H. S., Z. L. Wu, Y. M. Wu, S. M. Yu, D. N. Zhang, and W. H. Huang (2011). Developing a prototype earthquake early warning system in the Beijing Capital Region, *Seismol. Res. Lett.* **82,** 394–403.

Picozzi, M., D. Bindi, P. Brondi, D. Di Giacomo, S. Parolai, and A. Zollo (2017). Rapid determination of P wave-based energy magnitude: Insights on source parameter scaling of the 2016 Central Italy earthquake sequence, *Geophys. Res. Lett.* **44,** 4036–4045.

Satriano, C., L. Elia, C. Martino, M. Lancieri, A. Zollo, and G. Iannaccone (2011). PRESTo, the earthquake early warning system for southern Italy: Concepts, capabilities and future perspectives, *Soil Dynam. Earthq. Eng.* **31,** 137–153.

Satriano, C., Y. M. Wu, A. Zollo, and H. Kanamori (2011). Earthquake early warning: Concepts, methods and physical grounds, *Soil Dynam. Earthq. Eng.* **31,** 106–118.

Sesetyan, K., C. Zulfikar, M. Demircioglu, U. Hancilar, Y. Kamer, and M. Erdik (2011). Istanbul earthquake rapid response system: Methods and practices, *Soil Dynam. Earthq. Eng.* **31,** 170–180.

Sheen, D. H., J. H. Park, Y. J. Seong, I. S. Lim, and H. C. Chi (2016). Application of the maximum-likelihood location method to the earthquake early warning system in South Korea, *Bull. Seismol. Soc. Am.* **106,** 1064–1073, doi: 10.1785/0120150327.

Strauss, J. A., and R. M. Allen (2016). Benefits and costs of earthquake early warning, *Seismol. Res. Lett.* **87,** 765–772.

Sun, J. B., H. Yue, Z. K. Shen, L. H. Fang, Y. Zhan, and X. Y. Sun (2018). The 2017 Jiuzhaigou earthquake: A complicated event occurred in a young fault system, *Geophys. Res. Lett.* **45,** 2230–2240, doi: 10.1002/2017GL076421.

Tarantino, S., S. Colombelli, A. Emolo, and A. Zollo (2019). Quick determination of the earthquake focal mechanism from the azimuthal variation of the initial P-wave amplitude, *Seismol. Res. Lett.* **90,** 1642–1649.

Trendafiloski, G., M. Wyss, and P. Rosset (2011). Loss estimation module in the second generation software QLARM, in *Human Casualties in Earthquakes: Progress in Modeling and Mitigation*, R. Spence, E. So, and C. Scawthorn (Editors), Springer, Berlin, Germany, New York, New York, 95–106.

Wald, D. J. (2010). PAGER-rapid assessment of an earthquakes impact, in *Encyclopedia of Solid Earth Geophysics*, H. Gupta (Editor), Part 5, 243–245, Springer, Berlin and New York.

Wald, D. J., P. S. Earle, T. I. Allen, K. Jaiswal, K. Porter, and M. Hearne (2008). Development of the U.S. Geological Survey's PAGER system (Prompt Assessment of Global Earthquakes for Response), *The 14th World Conf. on Earthquake Engineering*, Beijing, China, 12–17 October 2008, available at https://earthquake.usgs.gov/static/lfs/data/pager/Wald_et_al_(2008)_14WCEE_PAGER.pdf (last accessed January 2019).

Wan, K. S., S. D. Ni, X. F. Zeng, and S. Paul (2009). Real-time seismology for the 05/12/2008 Wenchuan earthquake of China: A retrospective view, *Sci. China Earth Sci.* **52,** 155–165.

Wang, T. Y., X. Jin, Y. X. Wei, and Y. D. Huang (2017). Real-time numerical shake prediction and updating for earthquake early warning, *Earthq. Sci.* **30,** 251–267.

Wang, W. T., S. D. Ni, Y. Chen, and H. Kanamori (2009). Magnitude estimation for early warning applications using the initial part of P waves: A case study on the 2008 Wenchuan sequence, *Geophys. Res. Lett.* **36,** 287–295.

Wang, Z. J., and B. M. Zhao (2018). Method of accurate-fast magnitude estimation for earthquake early warning-trial and application for the 2008 Wenchuan earthquake, *Soil Dynam. Earthq. Eng.* **109,** 227–234.

Wells, D. L., and K. J. Coppersmith (1994). New empirical relationships among magnitude, rupture length, rupture width, rupture area, and surface displacement, *Bull. Seismol. Soc. Am.* **84,** 974–1002.

Wen, R. Z., and Y. F. Ren (2014). Strong-motion observations of the Lushan earthquake on 20 April 2013, *Seismol. Res. Lett.* **85,** 1043–1055.

Wu, C., H. Nakamura, and S. Horiuchi (2008). Rupture imaging of the Wenchuan earthquake and advice for Chinese earthquake early warning system, *The 14th World Conf. on Earthquake Engineering*, Beijing, China, 12–17 October 2008.

Wu, Y. M., and T. L. Teng (2002). A virtual subnetwork approach to earthquake early warning, *Bull. Seismol. Soc. Am.* **92,** 2008–2018.

Wu, Y. M., N. C. Hsiao, W. H. K. Lee, T. L. Teng, and T. C. Shin (2007). State of the art and progress in the earthquake early warning system in Taiwan, in *Earthquake Early Warning Systems*, P. Gasparini, G. Manfredi, and J. Zschau (Editors), Springer, Berlin, Germany, New York, New York, 283–306.

Wu, Z. L., C. S. Jiang, X. J. Li, G. J. Li, and Z. F. Ding (2014). *Earthquake Phenomenology from the Field: The April 20, 2013, Lushan Earthquake*, Springer, Singapore, Singapore, doi: 10.1007/978-981-4585-15-6.

Wyss, M. (2014). Ten years of real-time earthquake loss alerts, in *Earthquake Hazard, Risk, and Disasters*, M. Wyss (Editor), Elsevier, Waltham, Massachusetts, 143–165.

Wyss, M., and M. Zibzibadze (2009). Delay times of worldwide global earthquake alerts, *Nat. Hazards* **50,** 379–387.

Wyss, M., M. Elashvili, N. Jorjiashvili, and Z. Javakhishvili (2011). Uncertainties in teleseismic earthquake locations: Implications for real-time loss estimates, *Bull. Seismol. Soc. Am.* **101,** 1152–1161.

Wyss, M., A. Nekrasova, and V. Kossobokov (2012). Errors in expected human losses due to incorrect seismic hazard estimates, *Nat. Hazards* **62,** 927–935.

Wyss, M., R. J. Wang, J. Zschau, and Y. Xia (2006). Earthquake loss estimates in near real-time, *EOS Trans. AGU* **87,** 477–492.

Xu, X. W., X. Z. Wen, J. Q. Ye, B. Q. Ma, J. Chen, R. J. Zhou, H. L. He, Q. J. Tian, Y. L. He, Z. C. Wang, *et al.* (2008). The MS 8.0 Wenchuan earthquake surface ruptures and its seismogenic structure, *Seismol. Geol.* **30,** 597–629 (in Chinese with English abstract).

Yamada, M. (2014). Estimation of fault rupture extent using near-source records for earthquake early warning, in *Early Warning for Geological Disasters*, F. Wenzel and J. Zschau (Editors), Springer, Berlin, Germany, New York, New York, 29–47.

Yamada, M., and T. H. Heaton (2008). Real-time estimation of fault rupture extent using envelopes of acceleration, *Bull. Seismol. Soc. Am.* **98,** 607–619.

Zheng, G. G. (2018). Decade review and prospective of seismic risk reduction in China, *Keynote Lectures in International Conf. for the Decade Memory of the Wenchuan Earthquake with The 4th International Conf. on Continental Earthquakes (The 4th ICCE) and The 12th General Assembly of the Asian Seismological Commission (ASC)*, Chengdu, Sichuan, China, 12–14 May 2018, available at http://www.4thicce.com/ (last accessed September 2018).

Zhang, H. C., X. Jin, Y. X. Wei, J. Li, L. C. Kang, S. C. Wang, L. Z. Huang, and P. Q. Yu (2016). An earthquake early warning system in Fujian, China, *Bull. Seismol. Soc. Am.* **106,** 755–765, doi: 10.1785/0120150143.

Zhang, Y., R. J. Wang, Y. T. Chen, L. S. Xu, F. Du, M. P. Jin, H. W. Tu, and T. Dalm (2014). Kinematic rupture model and hypocenter relocation of the 2013 MW6.6 Lushan earthquake constrained by strong-motion and teleseismic data, *Seismol. Res. Lett.* **85,** 15–22.

Zhang, Y., R. J. Wang, J. Zschau, Y. T. Chen, S. Parolai, and T. Dahm (2014). Automatic imaging of earthquake rupture processes by iterative deconvolution and stacking of high-rate GPS and strong motion seismograms, *J. Geophys. Res.* **119,** 5633–5650, doi: 10.1002/2013JB010469.

Zhang, Y., G. H. Zhang, E. A. Hetland, X. J. Shan, H. P. Zhang, D. Z. Zhao, W. Y. Gong, and C. Y. Qu (2018). Source fault and slip distribution of the 2017 MW 6.5 Jiuzhaigou, China, earthquake and its tectonic implications, *Seismol. Res. Lett.* **89,** 1345–1353, doi: 10.1785/0220170255.

Zollo, A., G. Iannaccone, V. Convertito, L. Elia, I. Iervolino, M. Lancieri, A. Lomax, C. Martino, C. Satriano, E. Weber, and P. Gasparoni (2009). Earthquake early warning system in southern Italy: Methodologies and performance evaluation, *Geophys. Res. Lett.* **36,** L00B07, doi: 10.1029/2008GL036689.

Manuscript received 13 May 2019
Published online 28 January 2020